$\sqrt{(a^2-x^2)}$ =:

$7x = 1\sec\theta$

$7x = \sec\theta$

Quiz 11 :-

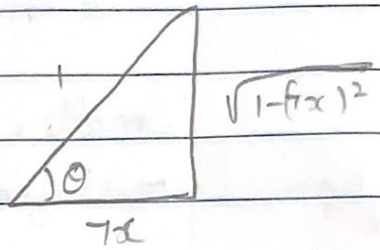
Q8:

(A) $\int x^2\sqrt{1-49x^2}\,dx = \int x^2\sqrt{1-(7x)^2}\,dx$

$7x = 1\sec\theta \quad \theta = \sec^{-1}(7x)$

$7x = \sec\theta$

$x = \frac{\sec\theta}{7}$



$dx = \frac{\sec\theta\tan\theta\,d\theta}{7}$

$= \int \left(\frac{\sec^2\theta}{49}\right)\sqrt{1-49\left(\frac{\sec^2\theta}{49}\right)}\left(\frac{\sec\theta\tan\theta\,d\theta}{7}\right)$

$= \int \sec^2\theta\tan\theta\,(\sec\theta\tan\theta\,d\theta)$

$= \int \sec^3\theta\tan^2\theta\,d\theta = \int \sec^2\theta\tan^2\theta\sec\theta\,d\theta$

$= \int (1-\tan^2\theta)\sec\theta\,(1+\tan^2\theta)\,d\theta$

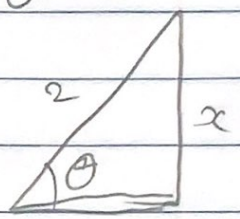
$= \int (7x)^2\left(\frac{1-49x^2}{49x^2}\right)(7x)\left(\frac{7\,dx}{(7x)\left(\frac{\sqrt{1-49x^2}}{7x}\right)}\right)$

$= \int \frac{49x\,(1-49x^2)}{\sqrt{1-49x^2}}\,dx$

$\frac{d}{dx}(\csc\theta) =$

(B) $\int \frac{\sqrt{x^2-4}}{x}\,dx \qquad \Rightarrow \quad x = 2\sin\theta$

$dx = 2\cos\theta\,d\theta$



$= \int \frac{\sqrt{4\sin^2\theta - 4}}{2\sin\theta}\, 2\cos\theta\,d\theta$

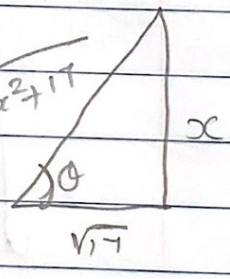
$= \int \frac{2\cos^2\theta}{\sin\theta}\,d\theta = 2\int \frac{1-\sin^2\theta}{\sin\theta}\,d\theta$

$$= 2\int \frac{1}{\sin\theta}\,d\theta - 2\int \sin\theta\,d\theta$$

(C) $\int \frac{1}{\sqrt{x^2+17}}\,dx$

$x = \sqrt{17}\tan\theta$

$dx = \sqrt{17}\sec^2\theta\,d\theta$



$= \int \frac{1}{\sqrt{17\tan^2\theta + 17}}\,\sqrt{17}\sec^2\theta\,d\theta$

$= \int \frac{1}{\sqrt{17}\sec\theta}\,\sqrt{17}\sec^2\theta\,d\theta$

$$= \int \sec\theta \, d\theta$$

$$= \int \frac{\sqrt{17}}{\sqrt{x^2+17}} \frac{dx}{\sqrt{17}\left(\frac{x^2+17}{17}\right)}$$

$$= \int \frac{17}{\left(\sqrt{x^2+17}\right)(x^2+17)} \, dx$$

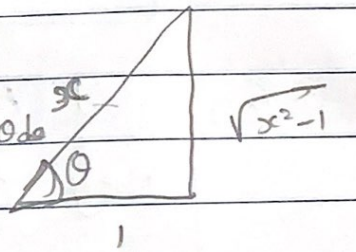
$$= \int \frac{17}{(x^2+17)^{3/2}} \, dx$$

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(2) $\int \frac{(x^2-1)^{3/2}}{x} \, dx$

$x = \sec\theta$

$dx = \sec\theta \tan\theta \, d\theta$



$$= \int \frac{(\sec^2\theta - 1)^{3/2}}{\sec\theta} \sec\theta \tan\theta \, d\theta$$

$$= \int \tan^3\theta \tan\theta \, d\theta$$

$$= \int \tan^2\theta (1 - \sec^2\theta) \, d\theta$$

Let $u = \tan\theta$, $du = \sec^2\theta \, d\theta$

$$= \int \tan^2\theta \, d\theta \; - \int u^2 \, du$$

$$= \int \tan^2\theta \, d\theta - \frac{u^3}{3} + C$$

$$= \int \tan^2\theta \, d\theta - \frac{\tan^3\theta}{3} + C$$

$$\Rightarrow \int 1 \, d\theta - \int \sec^2\theta \, d\theta - \frac{\tan^3\theta}{3} + C$$

$$= \theta - \tan\theta - \frac{\tan^3\theta}{3} + C$$

$$= \tan^{-1}\left(\frac{x}{\sqrt{17}}\right) - \frac{x}{\sqrt{17}} - \frac{x^3/(17)^{3/2}}{3} + C$$

$$= \tan^{-1}\left(x/\sqrt{17}\right) - \frac{x}{\sqrt{17}} - \frac{x^3}{51\sqrt{17}} + C$$

Q9:

$$\int \frac{2x-3}{x^3+x} \, dx = \int \frac{2x-3}{x(x^2+1)} \, dx$$

$$= \int \left(\frac{A}{x} + \frac{Bx+C}{x^2+1}\right) dx$$

$$A(x^2+1) + (Bx+C)x = 2x-3$$

$$Ax^2 + A + Bx^2 + Cx = 2x - 3$$

$\therefore$ $A + B = 0$ $\quad$ $\boxed{A = -3}$ $\quad$ $\boxed{B = 3}$

$\boxed{C = 2}$

$2\ln(x^2+1)$

$\frac{2}{(x^2+1)}$

$$= \int -\frac{3}{x} + \frac{3x+2}{(x^2+1)}\,dx$$

$$= -3\ln|x|\,dx + \int \frac{3x+2}{(x^2+1)}\,dx$$

$$\frac{3x+2}{x^2+1} = \frac{3x}{x^2+1} + \frac{2}{(x^2+1)}\,dx$$

$$= -3\ln|x|\,dx + \int \frac{3x}{x^2+1}\,dx + 2\ln|(x^2+1)|\,dx$$

$$= -3\ln|x|\,dx + \left(\frac{3}{2}\right)\ln|x^2+1| + 2\ln|x^2+1| + C$$

Let $u = x^2+1$

$du = 2x\,dx$

$\frac{3\,du}{2} = 3x\,dx$

(10)

(A) $$\int \tan^{-1}(x)\,dx = \int \frac{1}{\tan(x)}\,dx$$

$$= \ln|\tan(x)| + C$$

(B) $\int \frac{1}{x(\ln x + 2)^3} \, dx$

Let $u = \ln x + 2$ $du = \frac{1}{x} dx$

$= \int \frac{du}{u^3} = \int u^{-3} \, du$

$= \frac{u^{-2}}{-2} + C = \boxed{-\frac{1}{(\ln x + 2)^2} + C}$

(C) $\int (\sin^2 x + \cos^2 x) \, dx$

$= \int 1 \, dx$

$\boxed{= x + C}$